■ Block Production & Sales Performance Analysis

EXECUTIVE SUMMARY • PYTHON REBUILD & VALIDATION

■18.63M	■3.18M	17.1%
Reported revenue	Reported profit	Profit margin

Reporting period: December 2021 to December 2025
Business: Concrete block production and sales (6-inch, 7-inch, and 9-inch blocks)

1. Executive Overview

This project examines production, sales, profitability, customer behaviour, and credit risk for a block-moulding business. The analysis was originally developed through Excel dashboards and supporting SPSS/STATA statistical work, then rebuilt and extended in Python. The objective was to validate prior findings, correct identified data issues, and make the business implications easier to act on.

> **Headline insight:** production volume and profitability are not interchangeable. The reported analysis indicates that cost efficiency and sales realization matter more to profit than simply producing more blocks.

2. Operating Performance

Metric	Reported result
Total blocks produced	45,879
Total blocks sold	43,230
Sell-through	94.2%
Revenue	■18,633,540
Profit	■3,178,565
Profit margin	17.1%

These totals provide the scale of the business during the study window. The 94.2% sell-through figure is the reported ratio of blocks sold to blocks produced; it should be interpreted alongside stock movements because sales can be supplied from existing inventory.

3. What the Statistical Analysis Found

Production scale and profit

Production was strongly associated with sales, cost, and revenue (reported correlations around r = 0.82–0.85). Its relationship with profit was very weak (r = 0.09) and not statistically significant. This suggests that increasing output by

itself was not a reliable route to higher profit in these records.

Profit drivers

The reported OLS model identified sales volume and cost as important profit-related variables. The estimates were approximately ■221 higher monthly profit per additional block sold and about ■0.63 lower profit per additional ■1 in cost. Both estimates were reported as statistically significant at $p < 0.001$; the model R^2 was 0.49. A product-mix robustness check was also reported. These are model associations, not guaranteed marginal outcomes or causal effects.

Time trends and operational interruption

Revenue did not exhibit a statistically significant time trend ($p = 0.33$). Production declined by roughly 31 blocks per month ($p = 0.004$), with the earlier analysis attributing the interruption to an operational disruption. Sales continued during the low-production period, consistent with sales being supported by existing stock. The interpretation should be checked against the business's inventory and operational records.

4. Customers, Costs, and Credit Risk

Customer concentration

The reported customer analysis found that 40% of customers were repeat buyers and those customers generated 82% of revenue. This indicates that repeat relationships were central to revenue during the period. It also means that changes in the purchasing behaviour of a relatively concentrated group could materially affect sales.

Cost structure

The reported cost mix was 70.5% variable and 29.5% fixed. Sand and cement together accounted for roughly two-thirds of total cost. Cost per block increased from approximately ■148.8 in 2021 to ■546.7 in 2025. This progression makes unit-cost tracking and input-price monitoring important; the business should compare unit cost with realized selling price and product mix over time.

Bad debt

Bad debt was reported as 0.99% of billed amount overall. Its correlation with revenue was weak (r = 0.37) and not statistically significant in the available analysis. The overall rate appears limited, but customer-level balances and collection timing should be monitored before drawing conclusions about credit policy.

5. Data Quality and Validation

The Python workflow included checks against the earlier Excel and SPSS/STATA analysis. Nine of 11 cross-checked figures reportedly matched exactly, including correlation coefficients, the cost-per-block progression, a 51% pricing increase from 2023 to 2024, and both regression R^2 values. The notebook documents the comparison and two small narrative corrections.

• **Sales dates:** two records entered as 2014 were corrected to 2024 after review.

• **Customer pivot:** two customers whose sales were entirely recorded as bad debt were excluded from an earlier customer-value pivot; this understated total quantity sold by 63 blocks.

• **Monthly cost:** one blank total-cost value was derived from its single recorded cost component.

• **Missing production data:** blank values were preserved as missing where appropriate and were not automatically replaced with zero.

6. Business Implications

• **Track profit per block and cost per block.** Use unit economics—not output volume alone—to evaluate performance.

• **Pair production with sales and inventory.** When sales exceed current production, review stock drawdown and replenishment needs.

• **Prioritize input-cost visibility.** Monitor sand and cement prices and their effect on unit cost and margins.

• **Protect repeat-customer relationships.** Review retention, purchasing frequency, and revenue concentration.

• **Keep credit monitoring customer-level.** Follow overdue balances and bad-debt exposure without over-interpreting the weak aggregate correlation.

7. Scope and Interpretation

This is an observational analysis of the business records for December 2021–December 2025. Regression and correlation results describe relationships within the available data; they do not establish causation. Some production fields were missing, and the notebook documents the handling used for each relevant analysis. Figures and interpretations in this report are based on the project's reported analysis and should be checked against the source workbook and notebook before external publication.

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